



CSCloudSolutions

SuperAdmin Manual

FinOps SaaS · CSCloudSolutions

Version 2.0 · July 2026

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SuperAdmin Manual — FinOps SaaS (CSCloudSolutions)

Version: 2.0 (detailed) **Language:** English **Last updated:** July 2026 **Audience:** CSCloudSolutions internal team only (SuperAdmin). Includes everything in the User Manual plus platform administration flows (tenant onboarding, customer onboarding, sales tracking).

How to use this manual

Each section explains **what** the feature is, **who** can use it (role and subscription tier), and the **step-by-step usage flow** — concrete buttons, URL paths, form fields, and expected results. If you're a non-technical user, feel free to jump straight to the section you need: each one is self-contained.

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1. Getting Started

1.1. Access and login

The platform is a B2B SaaS with **two ways to sign in**, depending on your organization's cloud provider.

If you use Azure, authentication is integrated with **Microsoft Entra ID** (Azure Active Directory):

1. Go to the platform URL.
2. Click **"Sign in with Microsoft"**.
3. Authenticate with your corporate account. The platform automatically recognizes your Azure tenant and identity.
4. **Demo Mode:** if you want to try the platform without connecting your real environment, choose one of the preconfigured demo profiles from the main screen — they come with realistic simulated data and metrics, so you can explore every module risk-free. Pick the cloud provider in the demo form, or link straight to it with [/demo?tier=business&provider=aws](#). There are four demo tenants per provider (one per tier); the AWS ones use the

same scale factors as the Azure ones so a side-by-side comparison is honest. One AWS demo account is deliberately shown in `ERROR` state, so the demo also covers what a sync failure looks like.

If the tenant is AWS, authentication is platform-native (email and password). See [section 13](#).

1.2. The onboarding wizard (first time)

If you're an administrator and this is your organization's first time using the platform, upon login the **Setup Wizard** (`/onboarding`) automatically opens — a 5-step linear wizard with a progress bar:

Step	What you do	Result
1. Welcome & Company Info	Confirm company name, primary cloud provider (Azure), display currency (USD/EUR/GBP), and timezone	Your initial preferences are saved
2. Connect Azure Subscription	Paste Client ID , Client Secret , and Azure Tenant ID from the Service Principal (generated with the PowerShell script provided by CSCloudSolutions) and click "Validate"	The system checks live whether the Service Principal has the minimum required roles; if any is missing, it shows exactly which one in red
3. Run First Data Sync	Click "Run Sync"	Fetches your first set of cost data from Azure (can take up to 60 seconds)
4. Create Your First Budget	Fill in name, monthly limit (\$), and alert threshold (%)	Your first active budget is created
5. Set Up Notifications	Click "Configure" (opens <code>/admin/notifications</code> in a new tab)	Add at least one channel (email, Slack, or Teams) to receive alerts

You can **skip** any step and return later — the wizard keeps appearing until you complete or skip all 5. Progress (%) is calculated as $(\text{completed} + \text{skipped}) / 5 \times 100$. If you prefer the advanced setup form instead of the wizard, `/admin/onboarding` remains available in parallel.

1.3. Onboarding a new customer (SuperAdmin flow)

If you're a CSCloudSolutions SuperAdmin onboarding a new tenant:

1. **Register the Tenant:** go to `/admin/tenants` → **Tenant Management** → enter the customer's Entra ID, business name, and initial Tier.
2. **Generate credentials:** once created in the database, go to `/admin/onboarding` → **Customer Onboarding**. The **Client ID** and **Client Secret** fields now appear, to paste the Service Principal credentials generated by the PowerShell script you shared with the customer.
3. **Tag commercial origin (optional):** in the expanded panel for each tenant in the Environments Directory, the **"Commercial Origin / Referred by"** field lets you note which salesperson sold or referred the client — only visible to SuperAdmins.

Azure roles the script assigns, by contracted tier:

Tier	Built-in roles	Custom role
Essential	Reader, Cost Management Reader, Monitoring Reader, Billing Reader	—
Professional	Essential + Tag Contributor	—
Business	Professional + Tag Contributor	VM Start/Stop/Restart/Deallocate + tags
Enterprise	Business + Tag Contributor	Business + delete disk/snapshot/NIC/Public IP/NSG

⚠ **The 4 Essential roles are the absolute minimum** for the Real Consumption page to show data. If `Cost Management Reader` or `Billing Reader` is missing, Azure silently returns 0 rows.

⚠ **EA/MCA subscriptions** (Enterprise Agreement / Microsoft Customer Agreement) additionally require the customer's Billing Admin to assign `Enrollment Reader` or `Billing Account Reader` to the Service Principal at the billing account scope — the script can't do this automatically, it must be coordinated with the customer.

Verifying permissions were correctly assigned, after the customer runs the script:

- **Quick option (recommended):** `GET /api/admin/check-sp-roles?tenantId=<tenant-id>` — returns how many subscriptions have all required roles ✅, which are partial ⚠, and which have none ❌, with the exact detail of what's missing in each.
- **Manual option:** in Azure Portal → Subscription → **Access control (IAM)** → **Role assignments** → filter by the App Registration `CSCloudSolutions-FinOps-Agent` and confirm the tier's roles appear.

Troubleshooting — "Real Consumption shows no data":

Error message	Cause	Fix
<code>NO_COST_PERMISSION</code>	Missing <code>Cost Management Reader</code>	Assign the role to the SP on that subscription
<code>NO_SUBSCRIPTION_ACCESS</code>	Missing <code>Reader</code>	Assign <code>Reader</code> or re-run the script
<code>SUBSCRIPTION_INACTIVE</code>	The subscription has no spend in the current month or the last 30 days	Verify it's the correct subscription
<code>NO_SUBSCRIPTIONS</code>	The SP sees no subscriptions	Assign <code>Reader</code> on at least one
<code>NO_CONSUMPTION</code>	Everything OK but no spend in the current cycle	Wait for the billing cycle to close or check another subscription

2. Roles and Permissions

The platform separates **two independent concepts** that combine but don't replace each other:

2.1. Role (what you can DO)

Role	Capability
SuperAdmin	Full administration of CSCloudSolutions: create tenants, configure payment gateways. Internal role, not for customers.
Owner	Tenant owner. Full access, including plan changes and billing.
Admin	Complete financial visibility, configuration changes, corrective actions (shutting down VMs, deleting resources).
Colaborador (Collaborator)	Sees financial intelligence and visibility; can suggest changes but doesn't manage billing or users.
Reader	Read-only in dashboards and reports. Cannot apply changes or view sensitive configuration.

2.2. Domain permissions (which pages you SEE)

Beyond the role, each user can be assigned one or more **domain permissions**, which determine which sidebar sections they see — independent of their role:

Permission	Who benefits	What it enables to view
FinOps	FinOps Analyst	Cost and recommendation pages
CloudAdmin	Cloud Administrator	Infrastructure execution/write pages
Security	Security Auditor	Compliance auditing, credentials, governance
ProductOwner	Project/product lead	Visibility by Cost Center or application

How they combine: a user with the **Reader** role and the **FinOps** permission can view FinOps pages, but can't modify or delete anything in them — the role still governs actions. Permission filtering is opt-in: if a user has no permission assigned, they see everything their role and tier allow (default behavior, no domain restriction). **Admin/Owner** roles always see everything their tier allows, regardless of assigned permissions. The **Dashboard** (/) and **Support** (/support) are always visible to everyone.

How to assign permissions: go to </admin/users> → select the user → toggle the permission switches in the table, or assign them when manually creating a new user. (Note: users bulk-imported from Entra ID have no permissions assigned by default — assign them afterward from the main table.)

⚠ Domain permissions **do not** enable write actions on the backend — that remains exclusively controlled by the role. A user with the **CloudAdmin** permission but **Reader** role can view those pages, but any write action will return a 403 error if their role doesn't authorize it.

3. Main Navigation

The left sidebar groups everything into 5 pillars (collapsible sections — the active page's section auto-expands):

- **Visibility** — dashboards, academy, FinOps maturity
- **Financial Intelligence** — billing, budgets, optimization
- **Cloud Cleanup** — zombie resources, TTL
- **Governance** — tags, policies, approvals
- **Administration** — users, configuration, billing, API

A **search bar** at the top of the sidebar lets you find pages by name or content without manually navigating categories.

3.1. "History" button — time evolution of any metric

On Dashboard, Commitment Discounts, Rightsizing, Anomalies, Budgets, and High Availability you'll find a **History** button in the top right corner. When opened:

1. Choose a date range (up to **1 year back**).
2. See the daily evolution of that page's metrics as a **line chart** and **table**.

The platform automatically saves a daily snapshot of every page (retention ~13 months) — you don't need to enable anything, it's already running in the background.

3.2. Your profile

Click your **avatar** (circle with your initial, top right) to open:

- **Full name** — editable via the pencil icon.
 - **Email and role** within the tenant (read-only).
 - **Display currency** — currency selector.
 - **Appearance** — Light / Dark / Automatic (follows the OS theme).
 - **Sign out**.
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4. Visibility Section

4.1. Dashboard (White Board)

Your executive landing panel. Summarizes Total Potential Savings, detected Zombie Resources, and Governance rating at a glance.

How to use it:

1. Select the period at the top (current month, last 3/12 months).
2. Filter by subscription if you have several connected.
3. Pin/unpin cards for what you want to always keep at hand (pin icon on each card).
4. Any card locked by your current tier appears blurred — click it to see an upgrade modal with the detail of what it unlocks.

4.2. Costs and Projection ([/intelligence/cost-projection](#) , Professional+ tier)

Combines two tools in one page:

- **Cost Histogram:** daily spending distribution, with a range selector from the last month up to **13 months back** (everything Azure Cost Management allows querying), cached in Redis for instant response.
- **Cost Projection:** calculate how much you'll spend in the future.
 1. The system takes the monthly average of the last 12 months as a base.
 2. You enter an **expected annual growth %** (negative values allowed to simulate an optimization/savings scenario).
 3. Choose the horizon: 3, 6, 12, or 24 months.
 4. The result shows: base average, equivalent monthly rate, and projected total, with a real vs. projected line chart.

There's a summary card of this on the Dashboard with a **"See full detail"** link.

4.3. Azure Advisor

Direct sync with Microsoft's native recommendations, classified into Cost, Security, and Operational Excellence. Recommendations display in the language you have active on the platform (not Azure's original language).

How to use it: filter by category, review each recommendation's potential savings impact, and apply it directly from the platform or dismiss it with a justification if it doesn't apply to your case.

4.4. FinOps Maturity

Interactive assessment that places your organization at Crawl / Walk / Run across 5 pillars (visibility, optimization, governance, intelligence, operations). Answer the questionnaire to get a score per pillar plus a personalized

improvement roadmap.

4.5. FinOps Academy

Interactive learning center on FinOps and Azure cost optimization — structured courses, videos, glossary, and downloadable resources.

⚠ **Mandatory gate:** new users must complete the Academy before accessing the rest of the platform (with a disclaimer explaining why). Progress is tracked **per user**, not per organization — every new person on the tenant has to complete it, and once completed it can't be re-run. Visible to all roles except SuperAdmin.

4.6-4.10. Historical Progress, TOP Expenses, Resources, Green FinOps, Captured Savings, Financial Leaks (Professional/Business+)

- **Historical Progress:** monthly trend charts, savings milestones reached, month-over-month and year-over-year comparisons. You can add manual annotations explaining a change (e.g., "migrated to reservations").
- **TOP Expenses:** ranking of your largest spending sources by service, subscription, or resource group, with drilldown to individual resource level.
- **Resources** (Business+): complete inventory — advanced filtering and search, bulk tagging, export.
- **Green FinOps** (Professional+): carbon footprint estimate by service, comparison of most efficient regions, sustainability recommendations.
- **Captured Savings** (Professional+): tracking of real savings already achieved via RIs, Savings Plans, and Hybrid Benefit, with ROI calculation.
- **Financial Leaks** (Professional+): identifies wasted money (abandoned resources, over-sizing, unnecessary redundancy) with a remediation calendar.

5. Financial Intelligence Section

5.1. Real Consumption (</intelligence/billing> , Essential+)

The real-time, detailed billing dashboard, straight from Azure.

How to use it:

1. Filter by period, subscription, and resource group.
2. Zoom into a specific service to see its breakdown.
3. Download the invoice or export data to Excel from the corresponding button.
4. Create threshold alerts from the same page (takes you to the Self-Service Alerts form with context pre-filled).

5.2. Budgets (</intelligence/budgets> , Essential+)

1. **Create a budget:** name, period (monthly/quarterly/annual), limit in \$.
2. **Alert threshold:** define at what % of the budget you want to be notified (e.g., 75%).
3. The system tracks automatically — see real consumption vs. budget in real time, with history of previous periods.

5.3. Cost Groups (</intelligence/cost-groups> , Business+)

Custom cost grouping according to your own business logic (by project, line of business, application, or environment).

1. Create a group with naming rules (what resources belong, by name pattern or tag).

2. Manually assign resources if adjustment is needed.
3. Clicking a group opens a modal with tabs: **Costs, Actions, Resources, Governance** — each with detail specific to that group.
4. Compare groups against each other from the list view.

5.4. Active Reservations — Commitment Discounts (</intelligence/commitments> , Enterprise+)

Besides overall coverage and utilization, the **Active Reservations** table replicates Azure's *Reservations* blade: Name, Status, Expiration, Scope, Type, Product, Region, Renewal, Quantity, and last-day and last-7-day utilization.

- Click the **Renewal** button → modal to **enable/disable auto-renewal** for that reservation. The change applies directly in Azure — requires tenant Admin/Owner role **and** [Reservations Contributor/Owner](#) permissions in Azure.
- Click any **utilization percentage** → modal with last-day / 7-day / 30-day detail and daily trend.

5.5. Savings Plan vs Reservation (</intelligence/commitment-simulator> , Professional+)

Simulate both purchase options with your real numbers and compare 1-year and 3-year savings before committing.

5.6. What-If Simulator (</intelligence/simulator> , Enterprise tier for save/compare)

Simulate the impact of scaling compute/storage, varying network traffic, or enabling Azure Hybrid Benefit on your current cost — **before applying the real change**.

Full flow:

1. Move the sliders: compute scale, storage scale, network traffic increase %, enable/disable AHB.
2. Click **"Run Simulation"** — see the breakdown: base cost, projected cost, delta, and % change.
3. **Save the scenario:** click **"Save current"** → give it a name and optional notes. The scenario is frozen with those numbers (it won't recalculate later even if your real base cost changes over time — this lets you compare saved scenarios from different moments consistently).
4. Repeat 2-3 times with different sliders to have several saved scenarios.
5. **Compare:** check 2 to 4 scenarios with the checkboxes → click **"Compare"** → a side-by-side modal opens with each scenario as a card, marking which is the baseline.
6. **Export:** choose the format (CSV, PDF, or Markdown) next to the download buttons — you can export a single scenario, all saved ones, or the full comparison with each one's delta against the baseline.
7. **Delete:** trash icon on each row (only the creator or an Admin/Owner can delete it).

The assumed cost mix is Azure-first: 60% of cost is compute, 25% storage, 15% network; AHB applies a flat 18% discount on the total if enabled.

5.7-5.29. Other Financial Intelligence modules

Module	Tier	What it's for and how to use it
Cost Center Budget	Enterprise	Define cost centers, assign resources (manual or bulk), and the system automatically allocates spending to generate internal departmental chargeback.
Network Analysis	Business	Bandwidth, gateway, load balancer, and public IP costs; identifies transfer spikes and idle IPs to decommission.
Hybrid Benefit (AHB)	Business	Shows which VMs could use Software Assurance licenses and how much you'd save by enabling it; tracks what's already using it.
AKS Control	Enterprise	Active nodes, real utilization vs. over-provisioning, cost per pod, auto-scaling recommendations.
AKS Chargeback	Enterprise	Assign namespaces to teams and the system calculates how much each team spends on the cluster, for internal billing.
Container Apps	Business	Azure Container Apps cost control: monthly cost per app, environment, CPU/memory and replicas. Detects <i>scale-to-zero</i> opportunities (apps with a minimum replica ≥ 1 that could shut down when idle) and estimates the potential savings. Also available as a White Board card.
Log Analytics	Business	Log Analytics Workspaces cost control: monthly cost, retention and estimated ingestion per workspace. Detects unnecessary bulk ingestion (workspaces without a daily cap), excessive retention and <i>Commitment Tier</i> opportunities, with estimated potential savings. Also available as a White Board card.
Unit Economics	Enterprise	Define your own unit metric (cost per transaction, per user, per MB processed) and the system automatically calculates unit cost from your Azure data.
Users and Licenses (merged with "Licenses")	Professional	3 tabs: Dashboard (M365/Entra ID KPIs), User Activity (filterable table), License Optimization (resources without Hybrid Benefit via Resource Graph + per-SKU metrics via Microsoft Graph).
CSV Ingestion	Business	Upload a CSV with third-party billing under the FOCUS standard to analyze alongside your Azure data.
Self-Service Alerts	Business/Professional	Create budget or anomaly alert rules yourself, without asking support — condition + notification channel.
Scorecard	Business	0-100 score of your financial health with individual indicators (savings, reserve coverage, efficiency); you can set targets.
Tenant Health	Business	Overall status — detected issues, billing consistency, reservation coverage, policy compliance.
Rightsizing (+ VMSS/App Service/SQL/Storage verticals)	Enterprise	Analyzes 30 days of real usage and recommends the optimal SKU; shows estimated savings before approving the change. Dedicated verticals at /intelligence/rightsizing/{vmss, appservice, sqldb, storage} .
Storage Efficiency	Business	Simulates the savings from moving blobs between Hot/Cool/Archive tiers before applying it.
Compute \$/Core	Professional	Cost-per-vCPU breakdown to compare VM families against each other.
Anomaly Detection	Enterprise	Automatically identifies abnormal spending spikes or drops and lets you create alerts by adjusting sensitivity.

Module	Tier	What it's for and how to use it
Optimization Index (COIN)	Enterprise	Composite 0-100 score, comparable against industry benchmarks.
Compute Efficiency	Enterprise	Real CPU/memory/disk utilization per VM, with resize recommendation and calculated savings.
Rate Optimization	Enterprise	Compares your current rates against market benchmarks to prepare a negotiation with Microsoft.
Zero Cost	Everyone	Which resources are free in your subscription (free tier, credits) to maximize their use.
Allocation	Enterprise	Rules for distributing shared costs across multiple areas (by real usage, proportional, or fixed).
MACC Tracking	Enterprise	Tracking of the minimum annual commitment (EA/MCA) — consumed vs. committed, with compliance projection.
AI Cost Analytics	Enterprise	Cost per AI model and token consumption in Azure OpenAI, with call optimization recommendations.

6. Cloud Cleanup Section

6.1. Zombie Resources (`/cleanup/zombies` , Essential+; remediation Business+)

Detects orphaned resources generating unnecessary spend: unattached disks, unused public IPs, empty App Service Plans, VMs disconnected for 30+ days.

Usage flow:

1. List with filter by resource type.
2. Review each resource's last recorded use.
3. Optional: take a snapshot of the resource before touching anything (in case you need to recover it later).
4. **Delete** — requires Business+ role and Azure deletion permissions (see the script's role table in section 1.3).
5. You can create an **auto-cleanup policy** so zombie resources of a certain type get flagged or deleted automatically going forward.

6.2. Networking Zombies (`/cleanup/zombies/networking` , Essential+; remediation Business+)

Same as above but focused on network resources: empty Load Balancers, unassociated NSGs, orphaned Public IPs, VPN gateways with no active connections.

6.3. TTL Expirations (Business+)

Control of ephemeral environments (sandboxes, test environments) with an expiration date.

1. Create a TTL policy: what resource type, how many days of life.
2. Tag affected resources with the expiration date (manually or automatically by rule).
3. The system alerts you before automatic deletion.
4. Check the history of what was deleted and when.

7. Governance Section

7.1. Tag Compliance (</governance/tags> , Essential+; remediation Business+)

1. Define your organization's mandatory tags (e.g., [CostCenter](#) , [Owner](#) , [Environment](#)).
2. The system audits your entire infrastructure and shows you which resources lack them.
3. With **auto-tagging** (Business+) you can automatically apply missing tags based on rules.
4. Generate compliance reports to show internal audit.

7.2. Governance Reporting (</governance/reporting> , Enterprise+)

Unified executive dashboard: overall compliance status, security and tagging findings, prioritized recommendations, and historical evolution. (This page merges what used to be a separate "Governance Status" page as an additional section within the same report.)

7.3. Power Schedules (</governance/power> , Business+)

Automated VM power on/off routines outside productive hours.

Two modes:

- **Single date:** executes the action (power on/off/restart) **once** at the exact date and time.
- **Recurring (range):** define a **"From–To"** time range and the days of the week (e.g., Mon-Fri 08:00-20:00). The system automatically creates a power-on schedule at the "From" time and a power-off schedule at the "To" time, with the same days.

Important operational details:

- The timezone is auto-detected from your browser when you open the form — you can change it manually if you need a different one.
- The system checks pending schedules every **2 minutes**, plus an immediate check when you save. The VM action can take an additional 20-40 seconds to confirm against Azure.

7.4. High Availability (</governance/ha> , Business+)

Detects production VMs without an assigned Availability Zone or Availability Set — single point of failure risk.

7.5. Expiring Credentials (</governance/credentials> , Business+)

Proactive alert for App Registrations / Service Principals whose secrets or certificates expire in 30/60/90 days. Each credential shows status: **Expired**, **Expiring soon** (≤ 30 days), or **Enabled**.

How to create an alert:

1. **"Create expiration alert"** button.
2. Define how many days in advance you want the notice (1-365).
3. Choose the channel: email, Slack, or Teams (via webhook).
4. The system evaluates daily and sends **at most one notification per day** while there are credentials within the threshold (including already-expired ones).

These rules can also be managed from **Self-Service Alerts**, under the "Credential expiration" type.

7.6. Auto-Block Policies (Enterprise+)

Automatic policies that block actions before they happen: creating VMs above a certain size, creating resources without a mandatory tag, exceeding a maximum daily spend per subscription, or creating resources outside an allowed schedule.

7.7. Approvals (Business+)

Approval flow for infrastructure changes: a user requests the change, a specialist reviews and approves/rejects with comments, and everything is audited (who approved what and when).

8. Administration Section

8.1. Support ([/support](#) , all plans from Essential)

Any tenant user can open tickets to CSCloudSolutions and follow the conversation within the platform.

How to create a ticket:

1. Fill in subject, category (Technical / Billing / Inquiry / Feature Request), priority, and initial message.
2. You can attach screenshots or files ([jpg](#) , [jpeg](#) , [png](#) , [txt](#) , [json](#) ; max 5 MB per file, 10 per ticket) — kept for 60 days and then auto-deleted.
3. Support team replies appear marked with 🚑 in the thread. You can reply while it's open, and close/reopen it yourself.
4. **Quick access:** the lifebuoy icon next to your username in the header opens Support from any page. When your ticket gets a reply, you'll see a notification in the bell 🔔.

Quotas by plan:

Plan	Tickets/month	First-response SLA
Essential	5	48 h
Professional	20	24 h
Business	Unlimited	8 h
Enterprise	Unlimited	4 h

8.2. Users and Permissions ([/admin/users](#))

See section 2 for details on role vs. permissions. From here you add users, edit their role, toggle their domain permissions, or fully deactivate them.

8.3. Configuration ([/admin/config](#))

General tenant profile administration: name, logo, default language for new users, timezone for reports, billing cycle.

8.4. Billing — Plan Change ([/admin/billing](#) , Essential+, Owner role)

1. Choose the new plan (Essential / Professional / Business / Enterprise).
2. Choose frequency (monthly/annual) and proration mode.
3. The system shows a **preview summary** with the real amount calculated by the payment gateway before confirming: "You'll be charged \$X now" (upgrade) or "You'll receive a \$X credit" (downgrade), the new recurring total, and the next billing date.
4. The change **only applies** when you click **Confirm change** — until then you can cancel at no cost.

8.5. Notification Configuration ([/admin/notifications](#) , Professional+)

Three supported channels: **Slack**, **Microsoft Teams**, **Email (SMTP)**.

Configure Slack:

1. Go to <https://api.slack.com/apps> → create or select an app → **Incoming Webhooks**.
2. **"Add New Webhook to Workspace"** → choose the destination channel.
3. Copy the webhook URL (starts with <https://hooks.slack.com/services/...>).
4. In the platform: **Notifications** → **Add Channel** → **Slack** → paste the URL → save.

Configure Microsoft Teams:

1. In the Teams channel → ... → **Connectors** → search for "Incoming Webhook".
2. Configure the webhook and give it a name.
3. Copy the generated URL.
4. In the platform: **Notifications** → **Add Channel** → **Microsoft Teams** → paste the URL → save.

Configure Email:

- Uses SMTP. If your organization doesn't specify its own server, the platform's default SMTP configuration is used.
- You can define multiple recipients and, optionally, override host/port/user/password per channel.

Severity filter: each channel can filter what it receives — **info**, **warning**, **error**, or combinations (e.g., **warning,error** to skip informational messages).

Test the channel: a test button on each configured channel — sends a test notification before you rely on it in production.

Tier availability: Professional allows 2+ channels; Enterprise allows unlimited channels + per-channel SMTP overrides.

If notifications aren't arriving, check in this order: channel enabled (**enabled = true**) → severity filter matches the alert level → you tested the channel from the UI → there's at least one channel configured for the tenant. The send log (with errors) is available for diagnosis.

8.6. Executive Report (Business+)

Automated generation of high-level periodic reports, designed for presenting to leadership — downloadable executive summary.

8.7-8.17. Other Administration modules

Module	Tier	Usage
Customer Onboarding	Admin	See section 1.3 — tenant onboarding and Service Principal credentials.
Azure Lighthouse Onboarding	Enterprise	Generation of ARM template for cross-tenant delegation, at /admin/onboarding/lighthouse .
AI Configuration	Professional	Enable/disable AI features, choose model, adjust detection sensitivity and what data is shared.
Invoicing Report	Enterprise	JSON/CSV/PBIT stub export with detail by billing profile, invoice section, and customer, at /admin/report .
Workbooks	Enterprise	Custom reports: own layout, charts, data tables, exportable to PDF and shareable via link.
Audit Trail	Professional	Complete log of who changed what and when, filterable by user/action/date, exportable for compliance.
MCP API Keys	Enterprise	API keys for MCP-style integrations — generate, rotate, and revoke.
Public API	Enterprise	Documented REST API (OpenAPI/Swagger), with rate limits and usage monitoring.
Power BI Templates	Enterprise	Downloadable .pbit templates, pre-connected to your SaaS data.
FOCUS 1.1 Export	Professional	Export your data in the standard FOCUS format, schedulable for automatic daily generation.
SSO SAML	Enterprise	Configure your Identity Provider, map attributes (roles, emails), test, and activate for the whole organization.
Partner Markup (CSP)	Enterprise	Configurable margins per service for CSP partners, automatically applied in billing.
M365 Copilot	Enterprise	Tenant configuration + assisted chat over your own FinOps data, at /admin/copilot-m365 .

9. FinOps Copilot (AI Assistant)

Floating icon in the bottom corner of the screen, available on Professional+.

- **Automatic context awareness:** the Copilot reads the content of the page you're on — you don't need to tell it which module you're in. When you open it, without typing anything, it generates an **executive report** of what's being shown: module context, key findings, savings opportunities prioritized by impact, risks, and a 7-day action plan.
- **Targeted questions:** besides the automatic report, you can ask it directly. E.g., in Budgets: *"Summarize the current state of our budgets"*.
- **Corrective actions:** with your prior authorization, it can guide you through deleting zombie resources or applying missing tags via automated scripts — it never executes anything without your confirmation.

10. Account Security (MFA)

TOTP-based two-factor authentication (Google Authenticator, Microsoft Authenticator, Authy, etc.), optional per user, required only for **sensitive operations** (deleting a tenant, canceling a subscription, changing billing configuration).

How to enable it:

1. Go to your profile → **Security Settings** → **"Enable 2FA"**.
2. The system shows you a QR code — scan it with your authenticator app.
3. Enter the 6-digit code shown by the app to confirm.
4. You receive **10 recovery codes** — download and store them somewhere safe. They're shown only once.

When it's requested: when you attempt an operation marked as sensitive, a modal appears asking for your 6-digit code (or a recovery code if you've lost access to the authenticator). Each recovery code is single-use; if you use up all 10, you must disable and re-enable 2FA to generate a new set.

If you lose access to your authenticator: use any of your 10 recovery codes to log in and reconfigure 2FA from scratch. If you've also lost the recovery codes, an Admin can force-disable your MFA (logged in the audit trail).

11. Advanced Features and Integrations

This section is for technical users (Cloud Admin, DevOps) who need to integrate the platform with other tools or configure advanced access.

11.1. SSO SAML (`/admin/sso` , Enterprise, via WorkOS)

Lets Enterprise customer users log in with their own Identity Provider (Okta, Auth0, AD FS) instead of — or alongside — Microsoft Entra ID.

Configuration per customer:

1. Go to **Admin** → **SSO SAML** (only visible on Enterprise tier).
2. Get the `workos_org_id` and `workos_connection_id` from the WorkOS dashboard (CSCloudSolutions creates the Organization/Connection there if they don't exist).
3. Fill in the form: **Domain** (e.g., `acme.com`), **WorkOS Organization ID**, **WorkOS Connection ID** → enable the **Enable SSO** toggle → save.
4. **"Generate Admin Portal"** button → opens a WorkOS link in a new tab; the customer's IT admin receives an email to configure their own IdP.
5. (Optional) **"Test SSO"** button → redirects you to the IdP to validate the login works before announcing it to end users.

How the end user logs in: on the login screen, they choose "Login with SSO" instead of "Sign in with Microsoft" → the system redirects them to their own IdP → after authenticating, they're automatically returned to the dashboard.

⚠ Only **one IdP connection per tenant** is supported by current design. SSO coexists with Microsoft Entra ID login (MSAL) without replacing it — both methods work simultaneously, so you don't lose existing access when enabling SSO. The SSO session lasts 12 hours.

Common errors: "SSO not configured" (platform credentials missing), "SSO not enabled for this tenant" (the toggle is off), "Missing workos_connection_id" (the portal hasn't been generated yet, or the customer hasn't configured their IdP).

11.2. Audit Trail — advanced usage ([/admin/audit](#) , Professional+)

Beyond the filterable log (by email, action type, status, and date range, 10 rows per page), you have 4 export formats: **CSV (current page)**, **Full filtered CSV (streaming)**, **JSON**, **NDJSON**.

Typical use cases:

- **Compliance audit (SOC2):** export the full quarterly range with the filtered export button.
- **Investigate an incident:** filter by the suspicious user's email + **FAILURE** status.
- **Operational monitoring:** filter by action type **DELETE** + a recent date range to see what was deleted.

⚠ The full export has a **100,000-row cap** — for larger datasets, narrow the date range and export in parts. Log retention is currently indefinite (no automatic purge yet).

11.3. Public REST API v1 ([/admin/api-keys](#) , Enterprise)

Read-only programmatic access to your data (costs, budgets, recommendations, anomalies) to connect BI tools or your own scripts.

How to generate and use it:

1. **Admin → Public API** → generate a new key, choosing only the **scopes** you need (**read:cost** , **read:resources** , **read:budgets** , **read:recommendations** , **read:anomalies**) — request the minimum necessary, e.g., **read:cost** if it's only to feed a BI dashboard.
2. Authenticate each request with the header **X-API-Key: pak_live_xxx** (or **Authorization: Bearer pak_live_xxx**).
3. Example endpoints: **GET /cost/summary?from=...&to=...&groupBy=service** , **GET /cost/timeseries?granularity=daily|monthly** , **GET /resources?type=...&limit=100** , **GET /budgets** , **GET /recommendations** , **GET /anomalies?severity=low|medium|high** .
4. Interactive documentation (Swagger) available at [/api/v1/docs](#) .

⚠ Default limit: **60 requests/minute** per key (**X-RateLimit-*** headers show your remaining quota; hitting the limit returns a 429 error). If you need more for a batch job, request it from Admin → Public API or via support (up to 1000 req/min). Monetary values always travel as **strings** (e.g., **"1234.56"**), not numbers — parse them as text to avoid losing precision.

11.4. Showback/Chargeback PDF Export ([/admin/report](#) , Business+)

Generate and email showback/chargeback invoices to each customer or internal cost center.

- **Download a single invoice:** from the billing report, select the customer/center and period → the PDF downloads directly.
- **Download all at once:** same flow without selecting a specific customer → a ZIP with all the period's invoices downloads.
- **Send via email:** fill in the recipient's email and name → it's automatically sent with the PDF attached via Microsoft 365 (requires CSCloudSolutions to have email sending configured for your tenant — if unavailable you'll see a clear error asking you to contact support).

Every send and download is logged in the audit trail.

11.5. FOCUS 1.1 Export — advanced usage

Beyond manual download from [/admin/focus-export](#) (see section 8.7), you can automate extraction with your own MCP API Key (generated at [/admin/mcp-keys](#)):

```
curl -H "Authorization: Bearer mcp_xxx" \
".../api/exports/focus?tenantId=...&format=ndjson"
```

Available formats: CSV, JSON, NDJSON. Cap of **500,000 rows** per export (default 100,000). Useful for automatically feeding external FinOps tools (Power BI, CloudHealth, etc.) without manual intervention.

11.6. Status Page

A public page (no login required) where you and your users can check at any time whether the platform is operational: </es/status> (or </en/status>, </status>). Shows the status of API, database, Azure sync, AI provider, and billing, plus an incident history. Useful to share with your team if something seems broken — before opening a ticket, check whether an incident is already reported there.

11.7. Paddle Billing — technical detail

The plan change (see section 8.4) has 3 possible proration modes when upgrading/downgrading:

- **Immediate proration:** the difference is charged or credited right now.
- **Next-cycle proration:** the full price change is charged only at the next renewal.
- **No immediate charge:** the plan change applies without generating any charge until the next cycle.

From </admin/billing> you also access the **payment management portal** (update card) and the **downloadable invoice history**.

11.8. 7-day free trial (self-service)

If you signed up yourself from the pricing page (not through a SuperAdmin's onboarding), your account starts with a **7-day trial** on the plan you chose. You'll see a trial status banner at the top of the platform:

-  Blue: 5 or more days remaining.
-  Yellow: between 3 and 4 days remaining.
-  Red: 2 days or fewer — this one can't be dismissed.

You can upgrade to a paid plan at any time from **Billing** — the trial immediately converts to an active subscription. If the trial expires without an upgrade, the account switches to limited access until you activate a paid plan.

12. Best Practices

- **Weekly review:** check the **Dashboard** and **Zombie Resources** at least once a week to catch financial leaks before they pile up.
- **Automate early:** enable **Power Schedules** on your Development/Testing environments as a first step — it's common to see 60% savings in idle compute hours with no other change.
- **Enforce tag compliance:** without consistent tags, the chargeback/showback module can't fairly distribute the monthly bill across teams — it's the foundation everything else relies on.
- **Use the What-If Simulator before committing:** before purchasing a Reservation or Savings Plan, simulate the scenario and save it — it gives you a concrete number to justify the decision to finance.
- **Set up at least one notification channel from day one** (Slack/Teams if your team already lives there, or email if you prefer simplicity) — budget alerts are useless if nobody sees them in time.

13. Multi-cloud: provider model and data lifecycle

This section is SuperAdmin-only: it describes how the platform decides which cloud provider each tenant has, what happens when the plan changes, and how to operate data deletion.

13.1. The provider model

Every tenant has a `Tenants.provider` column with three possible values:

Value	Meaning	Minimum tier
<code>azure</code>	Azure only. This is the default and the state of every pre-existing tenant.	Essential
<code>aws</code>	AWS only. Written by the email+password signup.	Essential
<code>both</code>	Azure and AWS at the same time.	Enterprise

Exclusivity is **enforced server-side**, not only in the UI: `assertProviderIngestable()` blocks ingestion of the provider the tenant is not entitled to at all four entry points (AWS account creation, Cost Explorer sync, CUR sync and Azure credential upload). A tenant tampering with the frontend cannot ingest the provider they are not paying for.

13.2. Identity: two authentication paths

- **Azure** → Microsoft Entra ID (MSAL). The `tenant_id` is the Entra tenant GUID.
- **AWS** → platform-native identity (email + password). The `tenant_id` is a **generated UUID**, and `Users.entra_oid` stays `NULL`.

A user is "local" **if and only if** they have a `password_hash`. Platform tokens are signed with HS256 and Entra's are RS256; the backend discriminates by algorithm and each branch requires its own, so no algorithm confusion is possible. Roles, permissions, MFA and user quotas behave identically on both paths.

Deployment requirement: without the `LOCAL_AUTH_SECRET` environment variable (32 characters minimum) the seven local auth endpoints return **503 on purpose**. It is fail-closed: we would rather AWS login not work than work with a weak secret.

13.3. What happens when a `both` tenant downgrades

The downgrade arrives via webhook (Paddle, Azure Marketplace or AWS Marketplace) or via the SuperAdmin PATCH. All four go through the **same single choke point**, `applyTierChange()`, so none of them can forget the side effect.

The policy is **reversible archiving with a grace window**, never immediate deletion:

Moment	What happens
T+0 (downgrade)	One provider is elected to be retained and the other is archived : ingestion stops, all data and credentials are kept. Recorded in <code>TenantProviderTransitions</code> with status <code>GRACE</code> .
T+0 ... T+90	The archived provider is read-only . The FOCUS export stays enabled even if the new tier does not include it (data portability; GDPR art. 20), still requiring an ADMIN/OWNER role.
T-30 and T-7	The cron sends an email and an in-app notification. It is idempotent: each milestone is stamped on the transition row.
T+90	The cron purges the archived provider's data in batches of 5,000 rows and writes a <code>PROVIDER_DATA_PURGED</code> entry to <code>ActionLogs</code> .

Automatic election of the retained provider. If nobody chooses, the winner is: (1) higher spend over the last 90 days, (2) more connected accounts, (3) `azure` as the final tiebreaker. The spend comparison uses exact decimal arithmetic, never floating point: the result decides which dataset gets deleted.

The tenant can flip the choice for the whole window from the in-app banner or via `POST /api/admin/provider-transition` (ADMIN/OWNER role). **Flipping does not reset the clock** — if it did, a tenant could alternate indefinitely and keep multi-cloud for free forever.

Returning to Enterprise before the deadline restores everything with no loss. That is the case that justifies the entire window: a downgrade caused by a declined card is reverted within hours.

13.4. Why 90 days and not something else

- **Do not delete on the spot:** the event arrives through an asynchronous webhook, with nobody available to confirm a mass deletion; and the historical series cannot be rebuilt (Cost Explorer retains 12-14 months, and the CUR depends on a customer bucket we do not control).
- **Do not retain forever:** `FocusLineItems` is at resource/hour grain — millions of rows per account per month.
- **90 days = one full quarterly close.** The real case is the customer who downgrades in January and in April needs the whole Q1 to close the books.

It can be tuned with `PROVIDER_ARCHIVE_RETENTION_DAYS`. The value is **clamped between 7 and 730 days** rather than rejected, because a webhook and a cron consume it: a mistyped variable must not be able to purge tomorrow nor retain forever.

13.5. Operations

- **Notification and purge cron:** `/api/cron/provider-archive-purge`, authenticated with `Authorization: Bearer $CRON_SECRET`. **If it is not in the crontab, nothing is ever notified or purged** — the policy is left half-applied and retention becomes silently infinite. Recommended frequency: daily.
- **Auditing:** every transition lands in `ActionLogs` (`PROVIDER_ARCHIVED`, `PROVIDER_ELECTION_CHANGED`, `PROVIDER_RESTORED`, `PROVIDER_DATA_PURGED`) and in `TenantProviderTransitions`.
- **Verifying the first real case:** before the first downgrade of a large customer, review the first `PROVIDER_DATA_PURGED` by hand. The purge is irreversible and the only safety net is the full MySQL backup — there is no per-provider selective backup.
- **Checking any tenant's state:** `GET /api/admin/provider-transition?tenantId=...` returns the archived provider, the purge date and the days remaining.

13.6. Panel scope per provider

The panel was born 100% Azure and most pages end up calling Azure Resource Manager. The sidebar is filtered by the active provider using an **allow-list of routes**: by default a page is Azure-only, and only those explicitly marked as agnostic (platform, users, SaaS billing, auditing, FOCUS export) or AWS show up while AWS is active.

It is deliberately restrictive: if someone adds a new page and forgets to classify it, it stays **hidden** for AWS instead of appearing broken. As pages get parameterized, they are added to that list.

73 of 119 pages are enabled for AWS today. The cost pages all follow the same pattern: the route asks which provider the tenant uses and, when it is not Azure, skips the live Azure Cost Management call and reads straight from the database. What remains excluded is whatever depends on Azure Resource Graph inventory or on services with no direct equivalent (Azure Policy, Defender for Cloud, Hybrid Benefit).

Troubleshooting: empty resource inventory and tag audit on AWS

Two causes, and neither produces an error message — the symptom is always an **empty list**, which support tends to read as "the account has nothing in it":

1. **The role lacks `tag:GetResources` / `tag:GetTagKeys`**. They were added to the template in this release: tenants onboarded before it must **re-run it**. That is the first thing to check.
2. **The account has resources, but no tags**. The Resource Groups Tagging API —the only AWS API that lists resources across every service in a single call— only returns resources with **at least one tag**. There is no way to list untagged ones without walking service by service.

That is also how to read the tag compliance score: the denominator is the **tagged** resources, not the whole account. A tenant with immature tagging can show a high score precisely because the few resources it did tag, it tagged well.

On AWS that page is **read-only**: no remediation is offered because it would require `tag:TagResources`, a write permission the role does not request. The resource-group and tag-inheritance blocks are hidden, not broken: AWS has no container equivalent to a resource group.

⚠ **Enabling a route for AWS is always two changes, not one.** Besides adding it to the allow-list you must give it a case in the demo data generator. Doing only the first **does not fail visibly**: the AWS demo tenant falls through to the generic payload and sees **Azure resources**. This already happened on two pages before it was caught.

Cost allocation on AWS (allocation, chargeback, unit economics, showback)

These four capabilities were blocked for AWS for a structural reason, not an integration one: the daily cost aggregate discarded tags, so two rows for the same day, region and service with different cost centres **overwrote each other** and all spend landed in "Unallocated". It was fixed with a migration adding the tag dimension to the unique key. It was safe for already-persisted Azure data because the Azure ingest does not populate that column.

A limit that remains and support must be able to answer: a tenant that connected **only Cost Explorer, no CUR**, will still get no cost-centre breakdown. This is neither a bug nor a permissions issue — Cost Explorer does not return resource tags. The fix is for the customer to configure the CUR. This is already flagged on the onboarding screen and in the user manual.

13.7. AWS account onboarding: permissions and requirements

The account onboarding screen generates a **least-privilege** template (CloudFormation, Terraform or AWS CLI) containing exactly the actions the platform actually invokes, all **read-only**: `sts:AssumeRole`, `ce:GetCostAndUsage`, the EC2 inventory (`ec2:DescribeInstances`, `DescribeVolumes`, `DescribeAddresses`, `DescribeSnapshots`), native budgets (`budgets:DescribeBudgets`, `budgets:ViewBudget`, scoped to the account's own budget ARN) and `s3:GetObject` / `s3:ListBucket` scoped to the customer's CUR bucket. A test asserts that **closed list** and rejects any write verb, so it cannot grow without a real call justifying it.

⚠ **Tenants onboarded before July 2026 must re-run the template.** The original version only granted `ec2:DescribeInstances`, yet the idle-resource inventory already called `DescribeVolumes`, `DescribeAddresses` and `DescribeSnapshots`. **The symptom is not a visible error**: those families fail with `AccessDenied`, get dropped, and the customer sees an incomplete cleanup list that reads as "you have nothing to optimise". When an AWS tenant shows zero orphaned volumes or snapshots, check how old the role is before assuming the account is clean.

Previously we suggested the managed policies `job-function/Billing`, `AmazonEC2ReadOnlyAccess` and `AmazonS3ReadOnlyAccess`. The latter grants read access to **every** bucket in the customer's account, which is disproportionate for reading a cost report and is routinely rejected by demanding security teams.

Deployment requirement: without the `AWS_PLATFORM_ACCOUNT_ID` variable (the 12-digit ID of our AWS account), the template endpoint returns **503 on purpose**. The fail-closed behaviour is deliberate: issuing a template with the wrong account ID would have the customer grant access to their billing data to an account that is not ours.

When new capabilities require additional permissions (Cost Optimization Hub, AWS-native forecasting, and so on), they must be added to the template **at that time**, not in advance.

13.8. Cost Explorer pricing and caching

The AWS Cost Explorer API **charges USD 0.01 per request**, and each pagination page counts as a separate request. Left unchecked, a tenant with several accounts and a self-refreshing dashboard can produce a Cost Explorer bill larger than the savings the tool finds for them.

Reads are therefore cached in Redis per account and date range:

- **24 hours** if the range has already closed (past days do not change, barring billing adjustments).
- **1 hour** if the range includes the current day, which AWS keeps updating several times a day.

The **Test connection** button bypasses the cache on purpose: its job is to verify the role works *right now*, not to return what was read hours ago. Deleting an account invalidates its cache automatically.

If Redis is unavailable the read still goes to AWS: the cache is never a single point of failure for reading costs.

Support and Contact

For any additional assistance, open a ticket from **Support** ([/support](#)) within the platform, or write to soporte@csccloudsolutions.com.ar.

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